

Amat 19: Finite Mathematics

Lecture 01: Logic

(Propositions and Truth Tables)

At the end of this session, you must be able to...

- distinguish simple from compound propositions
- translate propositions from words into symbols and vice versa
- construct a truth table for a given proposition

Knights and Knaves

Once upon a time, there was an island that has two kinds of inhabitants: knights, who always tell the truth, and their opposites, knaves, who always lie.

You encounter two people A and B. What kind are A and B if A says "B is a knight" and B says "The two of us are opposite types"?

Who is a knave? Knight?

What is logic?

"Logic is neither a science nor an art, but a dodge."

-Benjamin Jowett

- Introduced by Aristotle ("the Father of Logic")
- Basis of all mathematical reasoning
- Concerned with whether reasoning is correct
- Focuses on the relationship among statements and not on the content of any particular statement

e.g. (1) All mathematicians wear sandals.
(2) Anyone who wears sandals is an algebraist.
(3) Therefore, all mathematicians are algebraists.

- Systematic thought, communication, debate, law, mathematics and science

Propositions

- **Proposition** – a **declarative** sentence whose truth value, whether **true or false but not both**, can be determined

E.g. Which of these are propositions?

1. Philippines is an archipelagic country.
 2. A century is 10 years.
 3. What time is it?
 4. Noah is a really, really nice guy.
- Small letters (usually p, q, r, s) called **variables** are used to represent propositions.

Simple vs Compound Propositions

- A proposition is ***atomic or simple*** if it cannot be syntactically analyzed into smaller parts; it is usually represented by a single logical variable.
- A proposition is ***compound*** if it contains one or more logical *connectives* ("and", "or", "not", "if...then")

Break down into simple propositions:

1. The number 10 is odd and the number 7 is even.
2. If Maria passes AMAT 19, then she will find a good job.
3. Kendra is not hungry.

Translating Words into Symbols

- Let p and q be simple prepositions.

Symbol	Logical Operator	Logical Connective
--------	------------------	--------------------

$\sim p$		
----------	--	--

	<i>negation</i>	
--	-----------------	--

		<i>not p</i>
--	--	--------------

$p \wedge q$		
--------------	--	--

	<i>conjunction</i>	
--	--------------------	--

		<i>p and q</i>
--	--	----------------

$p \vee q$		
------------	--	--

	<i>disjunction</i>	
--	--------------------	--

		<i>p or q</i>
--	--	---------------

$p \rightarrow q$		
-------------------	--	--

	<i>conditional</i>	
--	--------------------	--

		<i>if p then q</i>
--	--	--------------------

$p \leftrightarrow q$		
-----------------------	--	--

	<i>biconditional</i>	
--	----------------------	--

		<i>p if and only if q</i>
--	--	---------------------------

$p \oplus q$		
--------------	--	--

	<i>exclusive or</i>	
--	---------------------	--

		<i>p xor q</i>
--	--	----------------

Translating Words into Symbols

The conditional connective $p \rightarrow q$ represents the following English constructs:

- if p then q
- p only if q
- q follows from p
- p is a sufficient condition for q
- q if p
- p implies q
- q whenever p
- q is a necessary condition for p .

The biconditional connective $p \leftrightarrow q$ represents the following English constructs:

- p if and only if q (often written p iff q)
- p and q imply each other
- p is a necessary and sufficient condition for q
- p and q are equivalent.

Translating Words into Symbols

Translate the following propositions (if they are) into symbols:

1. Izza likes popcorn and Fred likes chocolates.
2. The weather forecast does not say that it will rain on Sunday.
3. If Ana sings on 22 August 2022, it will rain.
4. It's either I will get a 3.0 in AMAT 19 or I will get a 5.0.
5. Daehan loves to eat ice cream or corndog.

Rules on Logical Operators

Rules to remember about logical operators/connectives:

1. The negation $\sim p$ is true if p is false. If p is true, $\sim p$ is false.
2. The conjunction $p \wedge q$ is true if both p and q are true. Otherwise, it is false.
3. The disjunction $p \vee q$ is true if at least one statement (p , q , or both) is true. It is false only if both statements are false.
4. The conditional $p \rightarrow q$ is false only when the premise p is true and the conclusion q is false. Otherwise, it is true.
5. The biconditional $p \leftrightarrow q$ is true if p and q have the same truth value, that is, either p and q both true or both are false.

Truth Tables

- A ***truth table*** is a table that prescribes the defining rule for a logical operation. That is, for each combination of truth values of the operands, the table gives the truth value of the expression formed by the operation and operands.

Truth table

- A mechanism used in determining truth value of propositions

p	p	$\sim p$
T	T	F
F	F	T

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Simple vs Compound Propositions

- Truth values of simple propositions may be difficult to ascertain. (e.g. All mathematicians wear sandals.)
- Truth values of compound propositions can be determined on the basis of the truth values of the simple propositions that compose the compound propositions.

Order of operations

- In an **unparenthesized** compound proposition using only the five standard operators $\sim$, $\wedge$, $\vee$, $\rightarrow$, and $\leftrightarrow$, the following order of precedence is typically used when evaluating a logical expression, at each level of precedence moving from left to right:
first $\sim$, then $\wedge$ and $\vee$, then $\rightarrow$, finally $\leftrightarrow$.
- **Parenthesized** expressions are evaluated proceeding from the innermost pair of parentheses outward, analogous to the evaluation of an arithmetic expression.

Number of Cases

- Note that if there are 2 statements, there are 4 (or 2^2) rows/cases in the table showing all possible cases.
- If there are 3 statements, there will be 8 (or 2^3) rows/cases.
- In general, if there are n statements, then there will be 2^n cases

Examples for Truth Tables

Suppose p and q are simple propositions. What is the truth value of the following compound propositions? (Use truth tables)

1. $\sim(p \vee q)$
2. $\sim(p \wedge \sim q) \vee p$
3. $p \vee q \wedge \sim p \rightarrow q$

Tautology, Contradiction, Contingency

- Tautology – proposition that is always true regardless of truth values of individual propositions that compose it
- Contradiction – proposition that is always false regardless of truth values of individual propositions that compose it
- Contingency – proposition that has both some true and some false values based on the truth values of individual propositions that compose it